Analysis of Religion, Party, & Abortion
Paper 4

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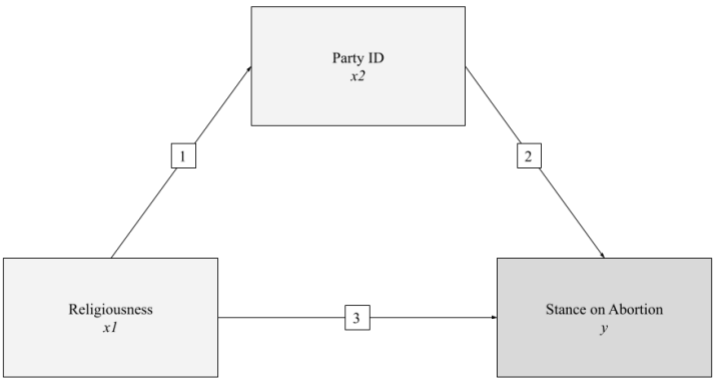
Theory



1. X_1 and X_2 = *Religiousness* on *Party ID*: The higher a respondent's reported strength of religion is, the more likely they will be to identify with the Republican Party given the Republican Party's use of religious rhetoric in politics and adherence to conservative Christian principles. The debate of abortion most often involves religious values on the conservative side, so it is likely that a more religious individual will resonate with the republican message.

2. X_2 and Y = *Party ID* on *Abortion Stance*: The more an individual identifies with the Republican Party, the more likely they are to not identify as prochoice, or be against abortion. Since the Republican Party traditionally follows the line of reasoning that life begins at conception from their interpretation of the bible, those who identify as Republican will be most likely to be against abortion. They will likely adhere to the party's opinion of abortion and therefore be more likely to express an anti-abortion stance and be against the pro-choice movement.

3. X_1 and Y = *Religiousness* on *Abortion Stance*: The higher a respondent's reported strength of religion is, the more likely they are to be against abortion and not identify as pro-choice. Those who are more strongly religious are more likely to accept strict interpretations of the bible, and believe that abortion is against their religious doctrine. Since the religious mainstream media is generally against abortion, and several political figures in the anti-abortion movement appeal to religion to garner anti-abortion support, those who are strongly religious are more likely to be swayed against the pro-choice stance.

Data

In GSS 2021, we use the relpersn, partyid, and prochoic variables. The relpersn variable asks respondent's to place themselbes on a four point scale of religiousness (1- very religious, 2- moderately religious, 3- slightly religious, 4-not religious at all). The partyid variable asks the respondent to identify their political party affiliation (0- strong democrat; 1-not very strong democrat; 2-independent, close to democrat; 3- independent; 4- independent, close to Republican; 5- not very strong republican; 6- strong republican; 7-other party). The prochoic variable asks respondents to what extent they agree with the statement "I consider myself pro-choice." (1 - strongly agree; 2 - agree; 3 - neither agree nor disagree; 4 - disagree; 5 - strongly disagree).The unit of analysis is American individuals. For the original frequency distribution, the relpersn variable shows 13.10% of respondents are very religious, 32.47% of respondents are moderately religious, 23.75% of respondents are slightly religious, and 30.68% of respondents are not religious at all. The highest categories of respondents' pecentage, then, is the moderately religious and not religious at all categories, while the lowest percentage of respondents identifies as very religious. For the partyid variable's frequency distribution, 20.550% identify as strongly democrat and 20.425% identify as identify as an independent (without sway) when excluding NAs. The lowest percentages of party affiliation for the respondents in this sample is 'other party' in category 7 at 2.850% and independents close to republican at 8.175%. For the frequency distribution of the prochoic variable, 29.65% strongly agree with the statement "I consider myself prochoice.", 28.27% agree, 20.53% neither agree nor disagree, 10.67% disagree and 10.87% strongly disagree with the statement. Therefore, in the original distribution, it can be seen that the highest percentages of respondents' answers occur in the agreement categories of 1 and 2, identifying with prochoice.(*Note all percentages referenced are valid percentages when NAs are excluded from the variable).

The independent variable, relpersn, will be recoded as religiousness with 3 categories instead of 4. The 4 category of 'not religious at all' will be recoded as 0, the 2 and 3 categories of 'moderately religious' and 'slightly religious' will be recoded as 1 meaning 'somewhat religious', and the 1 category of 'very religious' will be recoded as 2. Hence, the religiousness variable will follow a scale from 0 to 2, from 'not religious at all' to 'somewhat religious' then to 'very religious'. The frequency distribution for the recoded religiousness variable shows 30.68% of respondents identify as not religious at all, 56.22% of respondents identify as somewhat religious, and 13.10% of respondents identify as very religious. The sample clearly has the most respondents identifying with the somewhat religious middle category, and the least in the very religious. There are more not at all religious people in this sample than there are very religious people. The other dependent variable, prochoic, will be renamed prochoice and recoded with 3 categories instead of its original 4. For the purposes of the theory, the category 3 which corresponds to 'neither agree nor disagree' will be excluded as it does not express an opinion on the statement "I consider myself pro-choice." The recoded variable will recode category 5 as 0 to mean strongly disagree with statement, category 4 as 1 to mean disagree with statement, category 2 will remain as 2 to mean agree with statement, and category 1 as 3 to mean strongly agree with statement. The frequency distribution for prochoice shows 13.68% strongly disagree with the statement "I consider myself pro-choice.", 13.43% disagree with

the statement, 35.58% agree with the statement, and 37.31% strongly agree with the statement. The majority of individuals in the sample do identify as pro-choice as indicated by the agree category (2) and the strongly agree category (3). A roughly equal percentage of individuals disagree, and hence do not identify as pro-choice, including those strongly against the identification (category 0), with category 0 and 1 being around 13% of the respondents in the sample. Finally, the other independent variable, partyid, will be renamed as republican and consist of 3 categories: categories 0 (strong democrat) and 1 (not very strong democrat) will be recoded to 0 meaning democrat, categories 2-4 (independent close to democrat, independent, and independent close to republican, respectively) will be recoded as 1 meaning independent, and categories 5 (not very strong republican) and 6 (strong republican) will be recoded to 2 meaning republican, while category 7 with other party with be excluded as NA. The frequency distribution for republican shows 35.07% of respondents identify as democrat, 41.56% identify as independent, and 23.37% identify as republican.

```
# Step 2: Data


# load data
gss2021 <- haven::read_dta("gss2021.dta")

# keep only the relevant variables
gss <- gss2021[c("relpersn", "partyid", "prochoic")]

# labels, value labels, and frequency distribution as-is

list(attributes(gss$relpersn)$label, attributes(gss$relpersn)$labels,
    descr::freq(ordered(gss$relpersn), plot = F))

## [[1]]
## [1] "r consider self a religious person"
##
## [[2]]
##       very religious moderately religious   slightly religious
##                    1                    2                    3
## not religious at all
##                    4
##
## [[3]]
## ordered(gss$relpersn)
##       Frequency Percent Valid Percent Cum Percent
## 1           518  12.847         13.10       13.10
## 2          1284  31.845         32.47       45.57
## 3           939  23.289         23.75       69.32
## 4          1213  30.084         30.68      100.00
## NA's         78   1.935
## Total      4032 100.000        100.00

list(attributes(gss$partyid)$label, attributes(gss$partyid)$labels,
    descr::freq(ordered(gss$partyid), plot = F))

## [[1]]
## [1] "political party affiliation"
##
## [[2]]
##                    strong democrat           not very strong democrat
##                                  0                                  1
##     independent, close to democrat independent (neither, no response)
##                                  2                                  3
##   independent, close to republican         not very strong republican
##                                  4                                  5
```

```
##                  strong republican                        other party
##                                 6                                  7
##
## [[3]]
## ordered(gss$partyid)
##       Frequency  Percent Valid Percent Cum Percent
## 0           822  20.3869        20.550       20.55
## 1           541  13.4177        13.525       34.08
## 2           471  11.6815        11.775       45.85
## 3           817  20.2629        20.425       66.28
## 4           327   8.1101         8.175       74.45
## 5           384   9.5238         9.600       84.05
## 6           524  12.9960        13.100       97.15
## 7           114   2.8274         2.850      100.00
## NA's         32   0.7937
## Total      4032 100.0000       100.000
```

```
list(attributes(gss$prochoic)$label, attributes(gss$prochoic)$labels,
    descr::freq(ordered(gss$prochoic), plot = F))
```

```
## [[1]]
## [1] "i consider myself pro-choice."
##
## [[2]]
##             1.  strongly agree                      2.  agree
##                              1                              2
## 3.  neither agree nor disagree               4.  disagree, or
##                              3                              4
##         5.  strongly disagree?                     don’t know
##                              5                             NA
##                        refused                 skipped on web
##                             NA                             NA
##
## [[3]]
## ordered(gss$prochoic)
##       Frequency Percent Valid Percent Cum Percent
## 1          1053  26.116         29.65       29.65
## 2          1004  24.901         28.27       57.93
## 3           729  18.080         20.53       78.46
## 4           379   9.400         10.67       89.13
## 5           386   9.573         10.87      100.00
## NA's        481  11.930
## Total      4032 100.000        100.00
```

```
# recode, save it with a new name, show the updated
# frequency distribution

# category 0 as not religious, category 1 combines original
# categories 2 and 3 to represent somewhat religious, and
# category 2 represents very religious.
gss$religiousness <- car::recode(as.numeric(gss$relpersn), "1 = 2; c(2,3) = 1; 4 = 0")
descr::freq(ordered(gss$religiousness), plot = F)
```

```
## ordered(gss$religiousness)
##       Frequency Percent Valid Percent Cum Percent
## 0          1213  30.084         30.68       30.68
## 1          2223  55.134         56.22       86.90
## 2           518  12.847         13.10      100.00
## NA's         78   1.935
## Total      4032 100.000        100.00
```

```
# category 0 as not republican but democrat. category 1 as
# independent, category 2 as republican
gss$republican <- car::recode(as.numeric(gss$partyid), "c(0,1) = 0; c(2,3,4) = 1; c(5,6) = 2; 7 = NA")
descr::freq(ordered(gss$republican), plot = F)
```

```
## ordered(gss$republican)
##       Frequency Percent Valid Percent Cum Percent
```

```
## 0          1363  33.805         35.07       35.07
## 1          1615  40.055         41.56       76.63
## 2           908  22.520         23.37      100.00
## NA's        146   3.621
## Total      4032 100.000        100.00
```

```
# category 3 excluded since it outlines no opinion on the
# issue of abortion. Recoded category 5 as 0 to mean
# strongly disagree with statement 'I consider myself
# prochoice', category 4 to be 1 as disagree with
# statement, and category 1 to be 3 as strongly agree with
# statement. Category 2 will remain as 2.
gss$prochoice <- car::recode(as.numeric(gss$prochoic), "3 = NA; 5 = 0; 4 = 1; 1 = 3")
descr::freq(ordered(gss$prochoice), plot = F)
```

```
## ordered(gss$prochoice)
##       Frequency Percent Valid Percent Cum Percent
## 0           386   9.573         13.68       13.68
## 1           379   9.400         13.43       27.11
## 2          1004  24.901         35.58       62.69
## 3          1053  26.116         37.31      100.00
## NA's       1210  30.010
## Total      4032 100.000        100.00
```

Analysis

Hypotheses:

For religiousness and republican, I expect to find that as religiousness increases, so will republican. (Beta coefficient does not equal 0).

For religiousness and prochoice, I expect to find that as religiousness increases, prochoice will decrease. (Beta coefficient does not equal 0).

For republican and prochoice, I expect to find that as republican increases, prochoice will decrease. (Beta coefficient does not equal 0).

Bivariate Relationships

(1) Bivariate effect of religiousness X_1 on republican X_2

```
summary(lm(gss$republican ~ gss$religiousness))
```

```
##
## Call:
## lm(formula = gss$republican ~ gss$religiousness)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.15763 -0.69088  0.07575  0.30912  1.30912
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        0.69088    0.01969   35.09   <2e-16 ***
## gss$religiousness  0.23337    0.01891   12.34   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7428 on 3828 degrees of freedom
##   (202 observations deleted due to missingness)
## Multiple R-squared:  0.03828,    Adjusted R-squared:  0.03803
## F-statistic: 152.4 on 1 and 3828 DF,  p-value: < 2.2e-16
```

$$republican = 0.69088 + 0.23337 * religiousness$$

(2) Bivariate effect of republican X_2 on prochoice Y

```
summary(lm(gss$prochoice ~ gss$republican))

##
## Call:
## lm(formula = gss$prochoice ~ gss$republican)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4932 -0.4932  0.1317  0.5068  1.7565
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     2.49318    0.02531   98.52   <2e-16 ***
## gss$republican -0.62485    0.02221  -28.14   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8985 on 2721 degrees of freedom
##   (1309 observations deleted due to missingness)
## Multiple R-squared:  0.2254, Adjusted R-squared:  0.2251
## F-statistic: 791.7 on 1 and 2721 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.49318 - 0.62485 * republican$$

(3) Bivariate effect of religiousness X_1 on prochoice Y

```
summary(lm(gss$prochoice ~ gss$religiousness))

##
## Call:
## lm(formula = gss$prochoice ~ gss$religiousness)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.5370 -0.5370  0.1665  0.4630  1.8699
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)        2.53698    0.02756   92.07   <2e-16 ***
## gss$religiousness -0.70346    0.02648  -26.57   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9175 on 2813 degrees of freedom
##   (1217 observations deleted due to missingness)
## Multiple R-squared:  0.2006, Adjusted R-squared:  0.2003
## F-statistic: 705.9 on 1 and 2813 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.53698 - 0.70346 * religiousness$$

Bivariate Relationships Conclusion

Considering the regression equation between religiousness and republican, for each 1 category increase in strength of religiousness (from not religious at all to somewhat religious, or somewhat religious to very religious), there is, on average, a 0.23337 increase in categories of republican (0-democrat, 1-independent, 2- republican). Since a higher value for the republican variable corresponds away from democrat, the regression equation follows the theory that the more religious an individual is, the more likely they will be to identify as republican. At an alpha level of 0.05, the critical value for the t-statistic is 1.96.

Since the absolute value of the t-value for the coefficient is greater than the critical value, |12.34| > 1.96, the coefficient is statistically significant. We can reject the null hypothesis that no relationship exists between religiousness and republican. This is suggestive that individuals who are more religious are more likely to identify as republican, which is support for the causal theory between religiousness and party affiliation.

From the regression between republican and prochoice, for each one category increase in republican (from democrat to independent or independent to republican), on average, there will be a 0.62485 decrease in an individual's level of agreement with the statement "I consider myself prochoice." Since the prochoice variable is coded in such a way that lower categories indicate disagreement, this suggests that a more republican individual will be less likely, on average, to agree with the statement "I consider myself prochoice." The critical value for the t-statistic at an alpha level of 0.05 is 1.96. Since the absolute value of the t-value for the coefficient is greater than the critical value, |-28.14| > 1.96, there exists a statistically significant relationship between republican and prochoice. We can reject the null hypothesis that no relationship exists between republican and prochoice. In context, this follows the causal theory that republican individuals will be less likely to identify as prochoice.

From the regression equation between religiousness and prochoice, a one category increase in religious strength (from not religious at all to somewhat religious, or somewhat religious to very religious) will decrease an average respondent's agreement with the statement "I consider myself prochoice." by 0.70346 categories (0- strongly disagree, 1 - disagree, 2- agree, 3 - strongly agree). This suggests that as an individual's religious strength increases, they are less likely to be prochoice. The critical value for the t-statistic at an alpha level of 0.05 is 1.96. Since the absolute value of the t-value for the coefficient is greater than the critical value, |-26.57| > 1.96, the relationship is statistically significant. We can reject the null hypothesis that no relationship exists between religiousness and prochoice. This is suggestive of my theory that individuals who are more religious will be less likely to identify as prochoice.

There is a significant relationship between the two independent variables, as religiousness seems to affect party identification with the republicans; there is suggestive evidence that being more religious inclines an individual toward the republican party. Simultaneously, both independent variables, religiousness, and party affiliation, seem to also affect abortion stances. More religious respondents are less likely to identify with the prochoice movement, and republicans are also less likely to identify with the prochoice movement.

Conditional relationships

Hypothesis:

I expect that for Republicans, Independents and Democrats, as religiousness increases, identifying as prochoice will decrease. (Beta coefficient does not equal 0 for republicans, independents, and democrats).

(3) X_1 on Y given each category of X_2

When republican = 0 (democrats)

```
summary(lm(prochoice ~ religiousness, data = subset(gss, republican ==
    0)))

##
## Call:
## lm(formula = prochoice ~ religiousness, data = subset(gss, republican ==
##     0))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7357 -0.3416  0.2643  0.2643  1.0526
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.73568    0.02764   98.97   <2e-16 ***
## religiousness -0.39412    0.03056  -12.90   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6181 on 1077 degrees of freedom
##   (284 observations deleted due to missingness)
## Multiple R-squared:  0.1338, Adjusted R-squared:  0.133
## F-statistic: 166.4 on 1 and 1077 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.73568 - 0.39412 * religiousness$$

When republican = 1 (independents)

```
summary(lm(prochoice ~ religiousness, data = subset(gss, republican ==
    1)))

##
## Call:
## lm(formula = prochoice ~ religiousness, data = subset(gss, republican ==
##     1))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4503 -0.7702  0.2298  0.5497  1.9098
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.45025    0.04470   54.81   <2e-16 ***
## religiousness -0.68004    0.04339  -15.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9019 on 1008 degrees of freedom
##   (605 observations deleted due to missingness)
## Multiple R-squared:  0.1959, Adjusted R-squared:  0.1951
## F-statistic: 245.6 on 1 and 1008 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.45025 - 0.68004 * religiousness$$

When republican = 2 (republicans)

```
summary(lm(prochoice ~ religiousness, data = subset(gss, republican ==
    2)))

##
## Call:
## lm(formula = prochoice ~ religiousness, data = subset(gss, republican ==
##     2))
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8646 -0.6750 -0.2698  0.7302  2.3250
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.86459    0.08192  22.760   <2e-16 ***
## religiousness -0.59481    0.06553  -9.077   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9901 on 627 degrees of freedom
##   (279 observations deleted due to missingness)
## Multiple R-squared:  0.1161, Adjusted R-squared:  0.1147
## F-statistic: 82.38 on 1 and 627 DF,  p-value: < 2.2e-16
```

$$prochoice = 1.86459 - 0.59481 * religiousness$$

Conditional Relationships #1 Conclusion

For democrats, for each category increase in religiousness (not religious at all to somewhat religious, or somewhat religious to very religious) there is an average decrease of 0.39412 in agreement with the statement "I consider myself pro-choice." Considering the coding of the prochoice variable, a decrease would correspond to less agreement with the statement of self-identifying as pro-choice, or in favor of abortion. At the alpha level of 0.05, the value of the critical value for the t-statistic is 1.96, and the t-value of the coefficient is greater (|-12.90| > 1.96) than 1.96. Therefore, the relationship between religiousness and prochoice for democrats is statistically significant and we can reject the null hypothesis that there is no relationship between religiousness and identifying as prochoice for democrats. This relationship is suggestive that religiousness has an impact on abortion stance even for individuals who politically-identify with a party that is generally pro-choice, and also suggests that democrats with greater strength in religion are less likely to identify as pro-choice.

For independents, for each category increase in religiousness (not religious at all to somewhat religious, or somewhat religious to very religious), there is an average decrease of 0.68004 in agreement with the statement "I consider myself pro-choice." Considering the coding of the prochoice variable, a decrease would correspond to less agreement with the statement of self-identifying as pro-choice, or in favor of abortion. At the alpha level of 0.05, the value of the critical value for the t-statistic is 1.96, and the t-value of the coefficient is greater (|-15.67| > 1.96) than 1.96. Therefore, the relationship between religiousness and prochoice for independents is statistically significant and we can reject the null hypothesis that there is no relationship between religiousness and identifying with prochoice for Independents. This relationship is suggestive that religiousness still has an impact on those who fall outside of the main two-parties heavily engaged in the debate on abortion, and also suggests that independents with greater strength in religion are less likely to support abortion or identify as prochoice.

For republicans, for each category increase in religiousness (not religious at all to somewhat religious, or somewhat religious to very religious), there is an average decrease of 0.59481 categories in agreement with the statement "I consider myself pro-choice." Considering the coding of the pro-choice variable, a decrease would correspond to less

agreement with the statement. At the alpha level of 0.05, the value of the critical value for the t-statistics is 1.96, and the t-value of the coefficient is greater (|-9.077|>1.96) than 1.96. Therefore, the relationship between religiousness and prochoice for republicans is statistically significant and we can reject the null hypothesis that there is no relationship between religiousness and identifying as prochoice for republicans. This suggests that republicans with a greater strength in religion will be less likely to identify with the pro-choice stance on abortion.

The differences in the coefficient of the religiousness variable suggest the average impact of increases in religious strength on individuals in different parties. All coefficients for religiousness in every party are negative, suggesting there is a decrease in agreement with "I consider myself pro-choice." for every party when religiousness increases. Therefore, the impact of religiousness, on average, will decrease support/identification with pro-abortion stances. Comparing the magnitudes of the coefficient, independents have the greatest impact from religiousness on prochoice among all three parties (-0.68004). From this, it is suggested that independents, on average, exhibit the strongest aversion to pro-choice identification/pro-abortion stances when religiousness increases. Republicans also exhibit a strong decrease (-0.59481) per each category increase in religiousness. Democrats exhibit the lowest coefficient (-0.39412) per each category increase in religiousness. From this, it is suggested that independents are most influenced by religion in determining abortion stance followed by republicans. Democrats are suggested to be the least heavily influenced among the three party-groups in religiousness impacting abortion stance. The relationships, given they are all statistically significant, are also substantively significant. To elaborate, for democrats, increasing religiousness results in a third of a category decrease in identifying as pro-choice. For republicans and independents, there is over a half a category decrease in agreement with identifying as prochoice as religiousness increases, which is also substantively significant.

Hypotheses:

I expect that for non-religious, somewhat religious, and very religious individuals, as republican increases, identifying as prochoice will decrease. (Beta coefficient does not equal 0).

(2) X_2 on Y given each category of X_1

When religiousness = 0 (not religious at all)

```
summary(lm(prochoice ~ republican, data = subset(gss, religiousness ==
    0)))
```

```
##
## Call:
## lm(formula = prochoice ~ republican, data = subset(gss, religiousness ==
##     0))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7654 -0.3593  0.2346  0.2346  1.0469
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.76545    0.02956   93.57   <2e-16 ***
```

```
## republican  -0.40616    0.03333  -12.19   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6549 on 873 degrees of freedom
##   (338 observations deleted due to missingness)
## Multiple R-squared:  0.1454, Adjusted R-squared:  0.1444
## F-statistic: 148.5 on 1 and 873 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.76545 - 0.40616 * republican$$

When religiousness = 1 (somewhat religious)

```
summary(lm(prochoice ~ republican, data = subset(gss, religiousness ==
    1)))
```

```
##
## Call:
## lm(formula = prochoice ~ republican, data = subset(gss, religiousness ==
##     1))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.3311 -0.3311  0.1925  0.6689  1.7162
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.33109    0.03441   67.75   <2e-16 ***
## republican  -0.52364    0.02875  -18.21   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8752 on 1476 degrees of freedom
##   (745 observations deleted due to missingness)
## Multiple R-squared:  0.1835, Adjusted R-squared:  0.1829
## F-statistic: 331.6 on 1 and 1476 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.33109 - 0.52364 * republican$$

When religiousness = 2 (very religious)

```
summary(lm(prochoice ~ republican, data = subset(gss, religiousness ==
    2)))
```

```
##
## Call:
## lm(formula = prochoice ~ republican, data = subset(gss, religiousness ==
##     2))
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8329 -0.5712 -0.2020  0.7980  2.4288
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.83290    0.09298  19.713   <2e-16 ***
## republican  -0.63086    0.06611  -9.543   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9808 on 363 degrees of freedom
##   (153 observations deleted due to missingness)
## Multiple R-squared:  0.2005, Adjusted R-squared:  0.1983
## F-statistic: 91.06 on 1 and 363 DF,  p-value: < 2.2e-16
```

$$prochoice = 1.83290 - 0.63086 * republican$$

Conditional Relationships #2 Conclusion

Among non-religious voters, each category increase in republican (from democrat to independent, or independent to republican) will, on average, decrease agreement for the statement "I consider myself pro-choice." by 0.52364 categories as coded in the 4-category prochoice variable (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree). At an alpha level of 0.05, the critical value for the t-statistic is 1.96, and the t-value for the coefficient is greater in magnitude than this value (|-12.19|>1.96). Therefore, the result is statistically significant, and we can reject the null hypothesis that there is no relationship between republican (party affiliation) and identifying as prochoice for non-religious voters. This suggests that for non-religious individuals, affiliating with a more republican/less democratic party will prompt a lower likelihood of identifying as pro-choice.

Among somewhat religious voters, each category increase in republican (from democrat to independent, or independent to republican) will, on average, decrease agreement for the statement "I consider myself pro-choice." by 0.52364 categories as coded in the 4-category prochoice variable (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree). At an alpha level of 0.05, the critical value of the t-statistic is 1.96, and the t-value for the coefficient is greater in magnitude than this value (|-18.21| > 1.96). Therefore, the result is statistically significant, and we can reject the null hypothesis that there is no relationship between republican (party affiliation) and identifying as prochoice for somewhat religious voters. This suggests that for somewhat religious individuals, affiliating with a less democratic party/more republican party will lower the likelihood of identifying as pro-choice.

Among very religious voters, each category increase in republican (from democrat to independent, or independent to republican) will, on average, decrease agreement for the statement "I consider myself pro-choice." by 0.63086 categories as coded in the 4-category prochoice variable (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree). At an alpha level of 0.05, the critical value for the t-statistic is 1.96, and the t-value for the coefficient is greater in magnitude than this value (|-9.543|>1.96). Therefore, the result is statistically significant, and we can reject the null hypothesis that there is no relationship between republican (party affiliation) and identifying as prochoice. This suggests that for very religious individuals, affiliating with a 'more republican' party will lower the likelihood of identifying as pro-choice.

The differences in the coefficient of the republican variable, holding religiousness constant in each instance of the category, suggest the average impact of increases in level of republican on individuals with varying impacts of religiousness is consistently negative: being more religious in any political party decreases the likelihood of identifying with the pro-choice stance on abortion. Yet, when looking at the magnitude of these coefficients, it is suggested that very religious individuals, on average, exhibit the greatest aversion to the statement "I consider myself pro-choice" for each category increase in republican (from democrat to independent, or independent to republican) considering the coefficient for the very religious group is largest in magnitude (-0.63086). This suggests the greatest impact of party affiliation on abortion stance for the 'very religious' group in comparison to the 'somewhat religious' and 'not religious at all' groups. The smallest magnitude of all three

groups of religiousness is that of the 'not religious at all' group (-0.40616), which suggests that party has the weaker impact for the not religious at all individuals relative to the very religious and somewhat religious groups. This suggests that non-religious people experience the weakest effect of increasing republicanism on their abortion stance (their disagreement with identifying as pro-choice). The conditional comparisons may indicate an interaction between republican and religiousness is at play, considering the very religious category had the strongest aversion to identifying as prochoice, and the republican party individuals also had a strong aversion to identifying as prochoice as well, while we also have seen in the correlation matrix that there is a positive relevant correlation between religiousness and republican. Furthermore, the weakest decrease occurred in the democrat control and not religious at all control conditionals further justifying an examination for the opposite: republican and very religious.

Multivariate Regression

Correlation matrix

```
# drop NAs from all columns
gss_cor <- gss[, c("religiousness", "republican", "prochoice")] %>%
    drop_na(religiousness, republican, prochoice)

# correlation matrix
cor(gss_cor)

##               religiousness republican  prochoice
## religiousness     1.0000000  0.2483649 -0.4480148
## republican        0.2483649  1.0000000 -0.4742220
## prochoice        -0.4480148 -0.4742220  1.0000000

# correlation matrix = pearson coefficients of bivariate
# relationships
gss_cor$religiousness_std <- (gss_cor$religiousness - mean(gss_cor$religiousness))/sd(gss_cor$religiousnes
s)
gss_cor$republican_std <- (gss_cor$republican - mean(gss_cor$republican))/sd(gss_cor$republican)
gss_cor$prochoice_std <- (gss_cor$prochoice - mean(gss_cor$prochoice))/sd(gss_cor$prochoice)

summary(lm(gss_cor$prochoice_std ~ gss_cor$republican_std))

##
## Call:
## lm(formula = gss_cor$prochoice_std ~ gss_cor$republican_std)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4433 -0.4833  0.1283  0.4967  1.7199
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             3.643e-16  1.689e-02    0.00        1
## gss_cor$republican_std -4.742e-01  1.689e-02  -28.07   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8806 on 2716 degrees of freedom
## Multiple R-squared:  0.2249, Adjusted R-squared:  0.2246
## F-statistic:   788 on 1 and 2716 DF,  p-value: < 2.2e-16

summary(lm(gss_cor$prochoice_std ~ gss_cor$religiousness_std))

##
## Call:
## lm(formula = gss_cor$prochoice_std ~ gss_cor$religiousness_std)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.4938 -0.5338  0.1566  0.4462  1.8269
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                7.813e-16  1.715e-02    0.00        1
## gss_cor$religiousness_std -4.480e-01  1.715e-02  -26.12   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8942 on 2716 degrees of freedom
## Multiple R-squared:  0.2007, Adjusted R-squared:  0.2004
## F-statistic:   682 on 1 and 2716 DF,  p-value: < 2.2e-16
```

```
summary(lm(gss_cor$republican_std ~ gss_cor$religiousness_std))
```

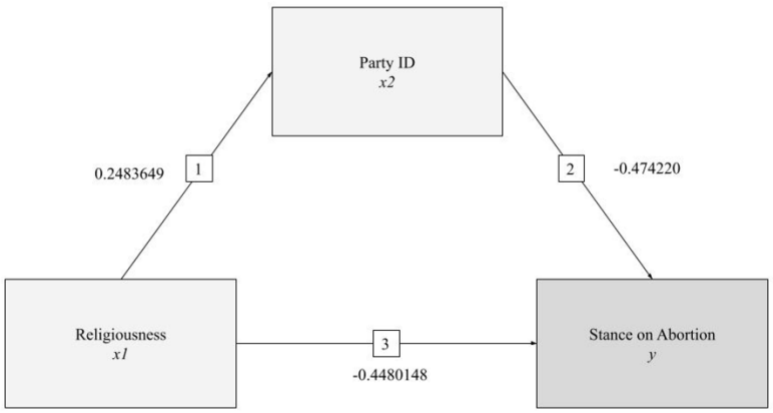
```
##
## Call:
## lm(formula = gss_cor$republican_std ~ gss_cor$religiousness_std)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.5307 -0.7653  0.1417  0.5244  1.8141
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)               -8.896e-16  1.858e-02    0.00        1
## gss_cor$religiousness_std  2.484e-01  1.859e-02   13.36   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9688 on 2716 degrees of freedom
## Multiple R-squared:  0.06169,    Adjusted R-squared:  0.06134
## F-statistic: 178.6 on 1 and 2716 DF,  p-value: < 2.2e-16
```

Correlation Matrix Analysis

The correlation matrix shows the Pearson correlation coefficients for each variable's relationship to each other, but they are standardized to compare each variable's units. Between republican and religiousness, one standard deviation increase in religiousness leads to a 0.2483649 standard deviation increase in republican. Republican and religiousness have a correlation coefficient of 0.2483649, showcasing a relevant correlation between increased strength of religion and identifying as closer to republican (since a magnitude of 1 indicates perfect correlation). Therefore, the positive correlation provides some support for the causal theory that religious individuals are more likely to be republican. Between religiousness and prochoice, one standard deviation increase in religiousness leads to a 0.4480148 standard deviation decrease in prochoice. Religiousness and prochoice has a correlation coefficient of -0.4480148, which suggests that there is a relevant negative correlation between being more religious and identifying oneself as pro-choice. The direction of the coefficient follows the causal theory that those who are more religious will be less likely to support abortion, and the magnitude of the coefficient showcases a relevant correlation between the two variables (religiousness and prochoice). Between republican and prochoice, for every-one standard deviation increase in republican, there is a 0.474220 standard deviation decrease in prochoice. The correlation coefficient between republican and pro-choice is -0.474220, which suggests

there is a relevant negative correlation between being republican and identifying as pro-choice. This follows the theory that republicans are less likely to support abortion, and hence less likely to be pro-choice. The correlation coefficients of religiousness and republican on pro-choice are very close in strength.

Path Diagram for Standardized Coefficients:



Multivariate Regression without interactions

```
summary(lm(gss$prochoice ~ gss$religiousness + gss$republican))

##
## Call:
## lm(formula = gss$prochoice ~ gss$religiousness + gss$republican)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8467 -0.3377  0.1533  0.6623  2.2781
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)         2.84670    0.02839  100.26   <2e-16 ***
## gss$religiousness -0.55337    0.02529  -21.88   <2e-16 ***
## gss$republican    -0.50904    0.02116  -24.05   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8286 on 2715 degrees of freedom
##   (1314 observations deleted due to missingness)
## Multiple R-squared:  0.3411, Adjusted R-squared:  0.3406
## F-statistic: 702.8 on 2 and 2715 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.84670 - 0.55337 \times religiousness - 0.50904 \times republican$$

The equation for the multivariate regression without interactions shows that for every one category increase in religiousness, there is a 0.55337 category decrease in

prochoice. In context of the variables, as religiousness increases from either 'not religious at all' to 'somewhat religious' or 'somewhat religious' to 'very religious', the level of agreement for the statement "I consider myself pro-choice." decreases by 0.55337 categories. This suggests that religious individuals are less likely to identify as prochoice. The equation also shows that for every category increase in republican (from democrat to independent or independent to republican), there is a 0.50904 category decrease in prochoice. In context of these variables, as republican increases a category, the level of agreement for the statement "I consider myself pro-choice." decreases by 0.50904 categories. This suggests individuals who identify as republican will be less likely to identify as prochoice. The critical value for the t-statistic at the alpha level of 0.05 is 1.96. For the religiousness coefficient, the t-value |-21.88| > 1.96, therefore the relationship between religiousness and prochoice is statistically significant. For the republican coefficient, the t-value |-24.05| > 1.96, therefore, the relationship between republican and prochoice is statistically significant. Again, this is suggestive evidence that individuals who are republican are less likely to identify as prochoice, and individuals who are religious are also less likely to identify as prochoice.

The coefficients change in the multivariate analysis as compared to the coefficients explored in the bivariate analysis for each independent variable's (republican and religiousness) effect on the dependent variable (prochoice -- level of agreement with the statement "I consider myself prochoice"). Since we looked at each one independent variable on the dependent in the bivariate regression, the coefficients were only taking into account either religiousness or republican (party affiliation). Hence, we got a coefficient of -0.62485 for each increase in republican (from democrat to independent, or independent to republican) on the level of agreement for identifying as prochoice in the 4-category variable prochoice (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree). In the multivariate regression, the coefficient for republican is -0.5094, interpreted the same, but this time with a smaller average 0.5094 decrease on level of agreement on identifying as prochoice for each category increase in republican. Similar effects are seen in the change on the variable religiousness on prochoice in the bivariate and multivariate regression without interaction. The coefficient for the bivariate regression shows that for each category increase in religiousness (from not religious at all to somewhat religious, or somewhat religious to very religious) there is a -0.70346 category decrease on the level of agreement for identifying as prochoice in the 4-category variable prochoice (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree). In the multivariate regression, the coefficient for religiousness is -0.55337, interpreted the same, but this time with a smaller average 0.55337 decrease on level of agreement on identifying as prochoice for each category increase in religiousness.

The coefficients have gotten smaller in the multivariate regression without interaction as compared to the individuals' bivariate regressions of each independent variable on the dependent variable, because we have now introduced both variables into the regression together. Therefore, for religiousness, we are now considering the impact of republican on prochoice also and vice versa. Since we are now controlling for each independent variable with the other, given that religiousness is associated with being more republican (as seen in the bivariate regression between religiousness and republican), we are now accounting for this affect, changing the coefficients. Moreover, in the bivariate

regression, only observations that did not include one independent variable (religiousness or republican) with the dependent (prochoice) were excluded for each regression. However, in the multivariate regression, observations without either of the independents and the dependent were excluded. This also has an effect on the calculation of the coefficients, changing them from the bivariate model.

Multivariate Regression with interactions

```
summary(lm(formula = gss$prochoice ~ gss$religiousness + gss$republican +
    gss$religiousness * gss$republican))
```

```
##
## Call:
## lm(formula = gss$prochoice ~ gss$religiousness + gss$republican +
##     gss$religiousness * gss$republican)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.7714 -0.3698  0.2027  0.6302  2.4182
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                       2.77144    0.03490  79.414  < 2e-16 ***
## gss$religiousness                -0.45176    0.03733 -12.101  < 2e-16 ***
## gss$republican                   -0.40164    0.03594 -11.175  < 2e-16 ***
## gss$religiousness:gss$republican -0.12070    0.03269  -3.693 0.000226 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8267 on 2714 degrees of freedom
##   (1314 observations deleted due to missingness)
## Multiple R-squared:  0.3444, Adjusted R-squared:  0.3437
## F-statistic: 475.2 on 3 and 2714 DF,  p-value: < 2.2e-16
```

$$prochoice = 2.77144 - 0.45176 \times religiousness - 0.40164 \times republican$$

$$-0.12070(religiousness \times republican)$$

Democrats: 2.77144 - 0.45176 x religiousness

Independents: 2.3698 - 0.5724 x religiousness

Republicans: 1.96816 - 0.70576 x religiousness

The coefficients of the bivariate regression equation show different impacts for different categories of religiousness and republican given the addition of the interaction term. For those who are democrat, the interaction term does not factor into the equation. This means for those who are democrats, for each category increase in religiousness (from 'not religious at all' to 'somewhat religious' or 'somewhat religious' to 'very religious'), there will be a 0.45176 category decrease in the 4-category level of agreement (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree) with the statement "I consider myself

pro-choice." The intercept for democrats, or average response in the case of non-religious democrats, is 2.77144.

For Independents, there is an additional 0.12070 category decrease along with the 0.45176 category decrease for each category increase in religiousness (from not religious at all to somewhat religious or somewhat religious to very religious). The total category decrease, on average, for independents in the 4-category level of agreement (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree) with the statement "I consider myself prochoice" is 0.5724 for each category increase in religiousness (not religious at all to somewhat religious or somewhat religious to very religious). The average level of agreement of independents who are not religious at all, is 2.3698 on the 4-category scale of agreement in the variable prochoice. This is lower than Democrats, possibly suggesting that since the Democratic Party adheres strongly to principles supporting pro-choice that Democrats would naturally be more likely than independents to identify as prochoice.

For Republicans, there is an additional 0.254 category decrease along with the 0.45176 category decrease in prochoice for each category increase in religiousness. The total average decrease in the 4-category level of agreement (0 - strongly disagree, 1 - disagree, 2 - agree, 3 - strongly agree) with the statement "I consider myself prochoice" is 0.70576 for each category increase in religiousness (not religious at all to somewhat religious or somewhat religious to very religious). The average level of categorical agreement with the statement "I consider myself prochoice" for non-religious republicans is the lowest among the three parties examined-- 1.96816. The average decrease per category increase in religiousness is evidently highest among Republicans, suggesting that strength of religion has the biggest negative impact on Republicans in identifying as prochoice. Moreover, even non-religious republicans have the lowest level of agreement in identifying as prochoice among all three parties. This is suggestive evidence for the theory that republicans will be least likely to support prochoice since they oftentimes follow religious principles in their politics.

The t-values of the coefficient, considering a critical value of 1.96 for the t-statistic at the alpha level 0.05, shows statistical significance for all coefficients, including the interaction coefficient. For the republican coefficient, $|-11.175| > 1.96$, for the religiousness coefficient, $|-12.101| > 1.96$, and for the interaction coefficient, $|-3.639|$ is statistically significant. Therefore, the relationships theorized about party affiliation, religiousness, and prochoice are supported by the regression (and more specifically the significance of the t-values of each coefficient). This is suggestive that strong religiousness and party affiliation toward republican decrease the level of support for identifying with prochoice. The interaction, as aforementioned, showcases that being very religious and republican makes one even less likely to be pro-choice. Finally, the relationship is also substantively significant, as any combination of religiousness and republican can impact the level of agreement with "I consider myself prochoice." by nearly half a category or more. Hence, the substantive and statistical significance acts as support for my causal theories that those who are more religious, and republican will be less likely to identify as prochoice.